# Project Proposal: An Auto-Research Loop with Self-Critique and Persistent Memory

## Overview

This proposal outlines a general-purpose research agent that accepts a research question, executes a structured investigation through code, web search, and dataset queries, and iteratively refines its conclusions by critiquing its own intermediate outputs. A persistent memory layer carries lessons across runs, allowing the system to avoid revisiting unproductive lines of inquiry. One sample financial application we intend to evaluate involves an alpha-discovery or macro-forecasting task, chosen so that the answer is not knowable in advance and the evaluation reflects genuine reasoning rather than recall.

## Motivation

Single-shot prompting captures only a narrow slice of what large language models can do when embedded in a deliberate workflow. A more faithful reflection of analytical practice involves hypothesis formation, evidence gathering, internal challenge, and revision. Existing agentic demonstrations typically lack two things: a critic that meaningfully disagrees with the first draft, and a durable record of prior attempts that informs subsequent ones. We propose to address both, and to measure whether the resulting system produces materially better answers than a one-shot baseline.

The contribution is methodological rather than purely empirical. By treating self-critique and memory as first-class components rather than prompt-engineering tricks, we aim to clarify when iterative refinement helps and, equally important, when it does not.

## System Design

The architecture consists of three coordinated components.

First, an iteration loop executes tool calls under PreToolUse and PostToolUse hooks, producing at each step a candidate answer paired with a structured rationale. Second, a critic stage operates as a separate pass with a distinct system prompt. Here, where feasible, a distinct model, is tasked with identifying weaknesses in the candidate answer and invoking tools independently to test them. If the critic surfaces a substantive flaw, the loop continues; otherwise the candidate is provisionally accepted. Third, a persistent memory layer records every (question, attempt, verdict) triple in a content-addressed log. On subsequent runs, a curated retrieval policy surfaces only the prior attempts most relevant to the current question, with the explicit goal of preventing repeated dead ends without overwhelming the agent's context.

## Data

The project will use real data exclusively; no synthetic fallback is planned. The primary evaluation question will be drawn from a public macroeconomic source, such as the Federal Reserve Economic Data (FRED) repository or the Bureau of Labor Statistics, paired with a research question for which no canonical answer exists. If broader coverage proves necessary, supplemental datasets from Kaggle and HuggingFace will be incorporated, with care taken to avoid problems whose tidy structure undermines the value of iterative reasoning.

## Evaluation

We will compare three system configurations on a held-out benchmark that is not shown to the agent during development:

• loop-N: the full system, with self-critique and persistent memory enabled.

• loop-1: a single critic pass, with memory disabled across runs.

• no-loop: a single-shot baseline without critique or memory.

The primary metric is task accuracy on the benchmark. Secondary measurements include iteration efficiency — the number of loops required to reach convergence — and a qualitative review of whether the critic identified substantive errors or merely generated superficially plausible objections. In keeping with the project's emphasis on methodological honesty, we will explicitly document at least one case in which self-critique degraded the answer rather than improving it. Such cases are informative about the limits of the approach and will be analyzed rather than excluded.

## Anticipated Risks

Several risks merit acknowledgment in advance. The critic may converge with the original reasoner when both share the same underlying model, producing agreement that reflects shared blind spots rather than genuine validation. We intend to mitigate this through aggressive variation of the critic's system prompt and, where practical, the use of a different model for the critic role.

Memory retrieval poses a second risk: over-retrieval dilutes the agent's context with irrelevant history, while under-retrieval reduces the memory layer to a vestigial component. The retrieval policy will require empirical tuning, and the parameters chosen will be reported transparently rather than obscured.

Finally, the iterative architecture is materially more expensive than single-shot inference. We will impose an explicit cap on iterations per run, log compute spend per configuration, and ensure that the cross-configuration comparison accounts for cost as well as accuracy.

## Deliverables

The project will produce a functioning agent and a reproducible repository.